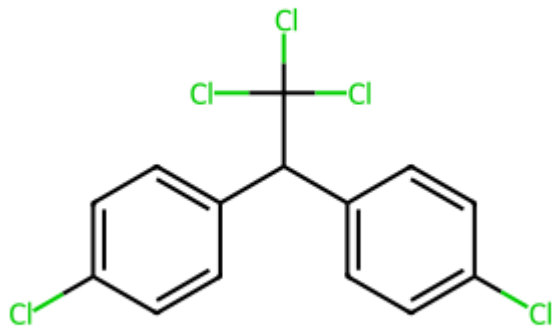
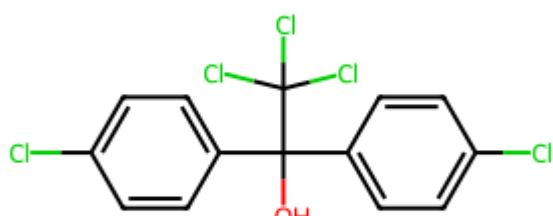




{'analysis_time': 0.0,
'first_solution_time': 19.976797,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 19.302838706963424,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 19.302838706963424,
'retro_star_value': 19.302838706963424}



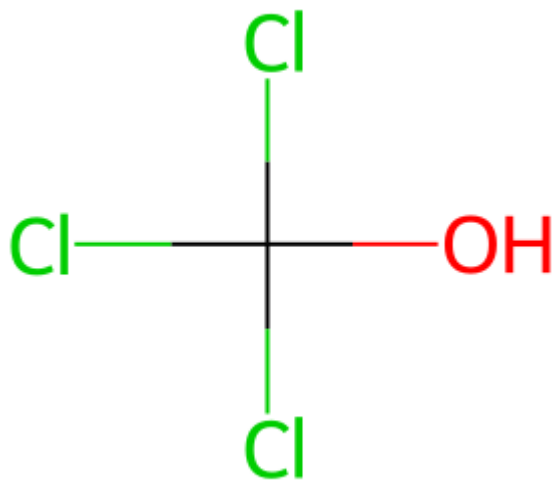
Reaction
{'analysis_time': 0.161517,
'first_solution_time': 46.467412,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 19.302838706963424,
'retro_star_reaction_number': 19.302838706963424,
'retro_star_rxn_cost': 0.14176053854627185,
'retro_star_value': 19.302838706963424}



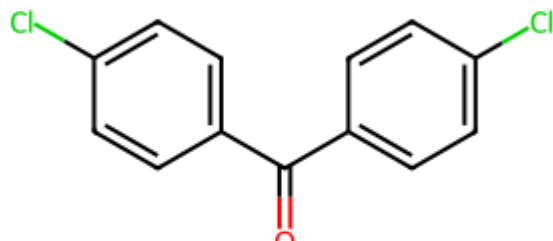
{'analysis_time': 0.161517,
'first_solution_time': 46.467412,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 19.161078168417152,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 19.161078168417152,
'retro_star_value': 19.302838706963424}



Reaction
{'analysis_time': 0.335357,
'first_solution_time': 46.467412,
'num_calls_rxn_model': 2,
'num_calls_value_function': 16,
'retro_star_min_cost': 19.161078168417152,
'retro_star_reaction_number': 19.161078168417152,
'retro_star_rxn_cost': 11.512925464970229,
'retro_star_value': 19.302838706963424}



{'analysis_time': 0.335357,
'first_solution_time': 46.467412,
'num_calls_rxn_model': 2,
'num_calls_value_function': 16,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 7.648152703446924,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 7.648152703446924,
'retro_star_value': 19.302838706963424}



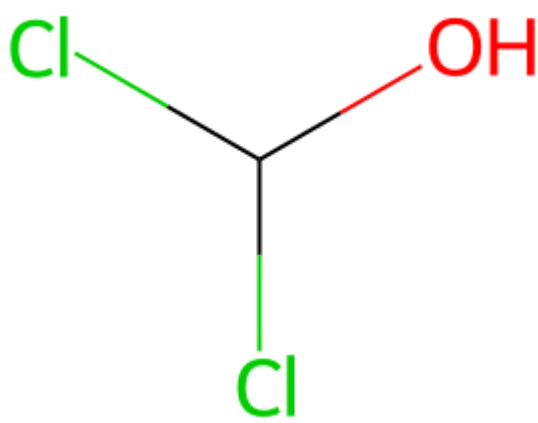
{'analysis_time': 0.335357,
'first_solution_time': 0.335357,
'num_calls_rxn_model': 2,
'num_calls_value_function': 16,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 19.302838706963424}



Reaction
{'analysis_time': 12.4124,
'first_solution_time': 50.269917,
'num_calls_rxn_model': 91,
'num_calls_value_function': 2978,
'retro_star_min_cost': 10.097590855963798,
'retro_star_reaction_number': 10.097590855963798,
'retro_star_rxn_cost': 5.0711857468813895,
'retro_star_value': 21.752276859480297}



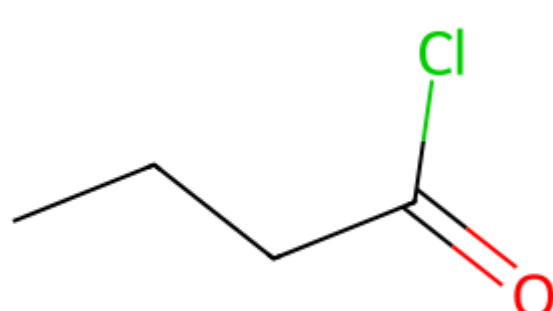
{'analysis_time': 12.4124,
'first_solution_time': 39.457674,
'num_calls_rxn_model': 91,
'num_calls_value_function': 2978,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 1.5304296406560596,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 1.5304296406560596,
'retro_star_value': 21.752276859480297}



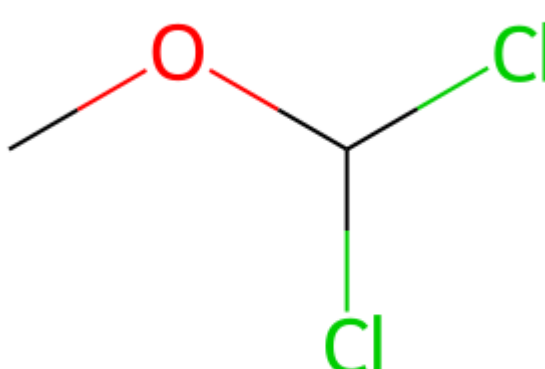
{'analysis_time': 12.4124,
'first_solution_time': 50.269917,
'num_calls_rxn_model': 91,
'num_calls_value_function': 2978,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 3.4959754684263493,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 3.4959754684263493,
'retro_star_value': 21.752276859480297}



Reaction
{'analysis_time': 39.457674,
'first_solution_time': 39.457674,
'num_calls_rxn_model': 261,
'num_calls_value_function': 7174,
'retro_star_min_cost': 6.848789537414928,
'retro_star_reaction_number': 6.848789537414928,
'retro_star_rxn_cost': 0.0,
'retro_star_value': 27.070636756239164}



{'analysis_time': 39.457674,
'first_solution_time': 39.457674,
'num_calls_rxn_model': 261,
'num_calls_value_function': 7174,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 27.070636756239164}



{'analysis_time': 50.269917,
'first_solution_time': 50.269917,
'num_calls_rxn_model': 338,
'num_calls_value_function': 9245,
'retro_star_min_cost': 5.80124122981574,
'retro_star_reaction_number': 5.80124122981574,
'retro_star_rxn_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 24.05754262086969}